CHAPTER 9

Combinatory and enumeration

S22, S25, S30, S35

Before you start · Sets and Choices

Without a calculator, in six minutes. Answer briefly. Mobilize trees of choice and factorial.
choose the targeted review indicated under the questions.

1. Count the exits of two coin throws.

2. Calculate 4 $\times 3 \times 2 \times 1$.

3. Distinguish "choose 2 students" and "designate president and then vice president".

4. How many choices for a vowel and then a consonant, with 5 vowels and 21 consonants?

5. Write $A \cap \overline{A}$.

6. List the parties to two elements of the $\{a, b, c\}$.

Responses

4; 24; the second choice is ordered; 105; $\varnothing$; $\{a, b\}, \{a, c\}, \{b, c\}$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Draw the four couples of results; R31. **2.** Perform multiplications from left to right. **3.** Compare (A,B) and (B,A) depending on the role of the students. **4.** Make two levels in the tree; R31. **5.** Find a which would satisfy both conditions. **6.** List without changing the writing order to avoid The questions R serve as support for the teacher, even if their collective ritual is planned later.

Objectives

Choose a counting model; distinguish order and repetition; use binomial coefficients and Pascal's relationship.

Course notes

Add or multiply

For finished sets two to two disjointed A_i, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, it is necessary correct double counting: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of components of the product Cartesian $A \times B$ is $|A||B|$. For a series of choices, multiply the number of possibilities at each stage, provided these numbers are valid for each previous choice considered.

Example. A two-letter code selected from among 26, followed by three digits, with authorized repetition, may take $26^2 \times 10^3$ values. A three-digit number does not start with 0: there are 9×10^2.

Lists and parts

In a set of n elements, there is n^k ordered lists of length k with repetition; $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ lists of k separate elements, if $0 \leq k \leq n$; $n!$ permutations of all We're laying $0! = 1$.

Combinations

A combination of k elements among n is a party to k elements; order does not intervene.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Why divide by $k!$? Each part of the size k gives exactly $k!$ separate lists according to order The number of lists without repetition is therefore $k!\binom{n}{k}$.

Identity and demonstration

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's relationship above is used for $0 \leq k < n$; formulas at the edge read directly with values 1.

Symmetry. Taking the supplement establishes a match between the size parts k and those Size $n-k$.

Pascal, combinatorial evidence. In a set of $n+1$ elements, fix an element a. Part of Size $k+1$ contains a (choose) k Other items n) or does not contain it (select $k+1$ elements among nBoth cases are disjunct and cover all possibilities.

Pascal, algebraic evidence. For $0 \leq k < n$, put at the same denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

The cases on the edge are dealt with directly.

Number of games. For each of the n items, choose "present" or "not present" gives 2^n parts. By classifying them by size, we count them $\sum_{k=0}^{n}\binom{n}{k}$: both counts are equal.

Deepening mentioned in the progression. Development $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ is understood by choosing, in each of the n factors, term a or b. This formula is not used as a prerequisite for the DS of this booklet.

Defining what it counts

9.1 Order and Repetition: Two Separate Questions

First use: S22.

Example resolved · Two letters selected from A, B, C

With order and repetition, the nine lists are AA, AB, AC, BA, BB, BC, CA, CB, CC.
Without repetition, six lists remain. Without order or repetition, the three objects are $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$. The result changes because the object counted changes, not because one chooses another formula.

Letter 1 / 2nd letter	A	B	C
A	AA	AB	AC
B	BA	BB	BC
C	CA	CB	CC

Barr the diagonal to prohibit repetitions; regroup AB and BA to forget the order.

A1 · Immediate application List the codes of two separate digits taken in $\{0, 1, 2\}$. How much are two-digit numbers ? Explain the difference between code and number.

Ownership · Addition and Product

The number of separate cases is by addition.
if each step offers the announced number of opportunities, taking into account previous choices.
For n separate objects, length lists k without repetition are number of $n(n-1)\cdots(n-k+1)$; full permutations are $n!$.

A2 · Partial guidance One outfit has one top among four and one pants among three. and pants are red. Count the outfits containing the two red pieces, then exactly A red room.

9.2 Forget the order without losing possibilities

First use: S30.

Definition · Binomial coefficient

$\binom{n}{k}$ is the number of parties to k elements of an assembly to n for $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. In particular $\binom{n}{0} = \binom{n}{n} = 1$.

Demonstration. Each party to k given elements $k!$ separate ordered lists. All parties give the same number of lists. $n!/(n-k)!$ lists without repetition by $k!$ therefore count each part exactly once. □

Example Solved · A Team and Roles

A team of three out of eight students can be selected from $\binom{8}{3} = 56$ ways. If one of the three becomes captain, there are $56 \times 3 = 168$ choices. $\binom{7}{2} = 168$ by first choosing the Captain. To assign three separate roles, there would be 8 $\times 7 \times 6 = 336$ possibilities.

A3 · Autonomous choice Ten students, including six seniors and four first students, form a team
Count the teams with exactly two finals, then at least one student of
each level, by two methods.

A4 · Repair Counting "Choose two delegates out of five gives 5 $\times 4 = 20$ possibilities."
Explain the error if the roles are identical. In what situation would the result 20 be correct?

9.3 Two proofs from Pascal, two readings of a sum

First use: S35 and S37.

Ownership · Demonstrated identities

For $1 \leq k \leq n$, $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$. In addition, $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Demonstration. Distinguish an element in a size set $n+1$. Size parts k who do not contain it are $\binom{n}{k}$; those which contain it require $k-1$ other elements: $\binom{n}{k-1}$. The two cases are disjointed and exhaustive. By calculation, put the two fractions at the same denominator $k!(n-k+1)!$ gives a numerator $n![(n-k+1)+k] = (n+1)!$. Finally, a part of a set n elements is built by n binary choices (take each element or not), i.e. 2^n parts; the classifying by size gives the sum. □

Solved Example · A Distinguished Element

Among five students, Lina, a team of two contains Lina (four choices for her partner) or does not contain it ($\binom{4}{2} = 6$ choices). $\binom{5}{2} = 4 + 6 = 10$.

A5 · Immediate application From line 1, 4, 6, 4, 1 of Pascal's triangle, build the
and verify the sum of its coefficients. Interpret coefficient 10.

9.4 Complementary and double counting

First use: S35.

Example resolved · At least one occurrence

A four-digit code in $\{0,1,2,3,4\}$ There's five of them. ^{4} codes and 4^4 codes without zero, therefore $5^4 - 4^4 = 369$ codes with at least one zero. Calculation 4×5^3 would count several times the codes with several zeros.

A6 · Choice of strategy Among these codes, count those that contain exactly two zeros.
count those whose sum of numbers is odd, without listing the 625 codes.

A first step: For the second question, classify according to the number of positions bearing an odd number: a or three.

Wait a minute at the courthouse. Explain why systematically divide a product by a factorial may be false: first specify what is considered to be the same result.

Review postponed · S41 and then S85. Find the binomial coefficient as number of paths having exactly the following: k success; compare this interpretation to that of a team.

Research activity

Order and repeat in a count · 30 min.

Six separate students volunteer to hold a debate.

1. We choose two students to form a pair without assigning a role.
   Can we train?

2. A president and a secretary are appointed, who must be two different students.
   Are both roles possible?

3. Three successive interventions are planned. The same student can intervene several times.
   programs of three interventions are possible?

4. We start again with only four pupils, appointed by A, B, C and D. Write all binomials
   and all the roles of the two roles.
   three interventions, without listing them.

5. For each of the three modes of choice, please indicate whether the order occurs and whether the same person can be selected several times.

Progressive exercises

E1 · Direct application. A code is a sequence of four digits chosen from 0, 1, . . . , 9.
Repetitions and the number 0 in the first position are allowed.
minus a 0?

E2 · Application and justification. Ten athletes take part in a race without ex æquo.
Different podiums, with a first, a second and a third place, can we get?

E3 · Method to be organised. Among seven distinct persons, a group of three persons is formed.
How many groups can be formed?

E4 · Choice of strategy. On a grid, we're going from $(0;0)$ to achieve $(4;3)$ by performing exactly four moves from one unit to the right and three to the top. How many paths
Can we follow each other?

E5 · Synthesis or demonstration. There are eight girls and seven boys in one class.
Four students with exactly two girls can be trained?

Work in the manual or annal

Barbazo, p. 294, exercise 37. At home after S35: deal with both questions. Specify if the order intervenes and if a card can be selected several times before using a binomial coefficient.

Duty at home

DM9 · Form a commission · 45 min, after S35.

A team consists of five teachers and seven students, all distinct.
Four people.

1. How many different commissions can be formed?

2. How many commissions contain exactly two students?

3. How many commissions contain at least one teacher and at least one student?

4. Among the members of a question 2 commission, a chair teacher is appointed and a chair teacher is appointed. student secretary. Count the choices of the commission accompanied by these two powers.

5. By counting the commissions according to their number of teachers, demonstrate that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$

QCM A · order and repeat

After S25; QCM B after S35.

For each question, only one proposal is accurate.
consult the correction.

1. A three-digit code selected from 0, 1, 2, 3, with authorized repetition, offers:

A. 4×3 B. 3^4

C. 4^3 D. 4!

2. Two distinct functions are assigned to two out of five persons.
is:

A. 10 B. 20

C. 25 D. 120

3. The number of permutations of four distinct objects is:

A. 16 B. 12

C. 4 D. 24

4. For two disjointed finite sets A, B, $|A \cup B|$ is valid for:

A. $|A| + |B|$ B. $|A||B|$

C. $|A| - |B|$ D. $|A \cap B|$

5. With two cards A and B, the lists of length 2 with repetition are:

A. AB, BA B. AA, BB

C. AA, AB, BA, BB D. A, B

QCM B · complementary and double counting

For each question, only one proposal is accurate.
consult the correction.

6. Choosing three out of eight students with no assigned function includes:

A. 8^3 B. $\binom{8}{3}$

C. $8 \times 7 \times 6$ D. 3^8

7. The number of parts of a five-element set is:

A. 5! B. 10

C. 25 D. 32

8. Pascal's relationship gives $\binom{6}{3}$ equal to:

A. $\binom{5}{2}+\binom{5}{3}$ B. $\binom{5}{2}\binom{5}{3}$

C. $2\binom{6}{2}$ D. $\binom{6}{4}+1$

9. One urn contains 3 distinct reds and 2 distinct blues.
red number of:

A. 6 B. 10

C. 9 D. 3

10. Six people are divided into two groups of three. The number of distributions is:

A. 10 B. 20

C. 40 D. 6!

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\binom{n}{0}=0$.

2. Counting by supplementary may avoid distinguishing several cases.

3. Shooting three objects simultaneously always gives an order to the objects.

Corrected MCQs

1. **C.** Three independent positions of four choices, including zero at the beginning: $4^3=64$.

2. **B.** $5\times4=20$. The order of duties counts and the same person does not cumulate both.

3. **D.** $4!=4\times3\times2\times1=24$.

4. **A.** Each element is counted once. Without disjunction, the intersection should be removed.

5. **C.** The list keeps the order and allows to reuse a card; there are $2^2=$ Four exits.

6. **B.** The group is not ordered, so $\binom{8}{3}=56$; the ordered product counts each group 3 ! Once.

7. **D.** For each element, two decisions: retained or not. $5^5=32$, including the empty part.

8. **A.** Separate groups containing a fixed element (choose two out of five) and those containing a fixed element Not containing (three out of five).

9. **C.** $\binom{5}{2}-\binom{2}{2}=10-1=9$. Other calculation: $3\times2+\binom{3}{2}=9$. Six only count mixed pairs.

10. **B.** Selecting the three members of Group 1 determines Group 2: $\binom{6}{3}=20$. Divide by 2 correct only for unnumbered groups.

True/false. False: there is an empty part, so $\binom{n}{0}=1$. True: Question 9. False: A simultaneous draw gives a part; add an order changes the object counted.

Corrected applications and support

Application 1. Codes: 01,02,10,12,20,21. The last four only represent numbers to
A code can start with zero.

Application 2. Two red pieces: one outfit. Exactly one: 1 $\times 2 + 3 \times 1 = 5$ outfits. Both cases are disjunct.

Implementation 3. Exactly two finals: $\binom{6}{2}\binom{4}{1} = 60$. Joint group: $\binom{10}{3} - \binom{6}{3} - \binom{4}{3} = 96$; or $\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$.

Implementation 4. Each unordered pair is counted twice; there is $\binom{5}{2} = 10$ pairs. roles distinguish the choices, hence 20.

Application 5. Line : 1, 5, 10, 10, 5, 1, sum 32. The first 10 count the parts of size 2 d size 5 set; the second size 3 set.

Implementation 6. Two zeros: $\binom{4}{2}4^2 = 96$. Of the five figures, three are even and two are odd. odd: $\binom{4}{1}2\,3^3 + \binom{4}{3}2^3\,3 = 216 + 96 = 312$.

Institutionalization

Ask before each calculation: "What is a possible result? When are two results
allow lists and trees in the small case, then make explicit what becomes general.
The calculation of parity can remain a challenge accompanied; the case-by-case method suffices.

Conduct and correction

Benchmarks for the teacher

The common duty S28 precedes the combinations of S30: to put only principles and lists
The DS3 can mobilize combinations, but not Pascal's relationship taught in
S35. Make the object counted before any formula. For fragile students, use cards
with names; does a card swap change the object? The Newton binome, quoted
in the progression, is a deepening: keep it explicitly optional.

Corrected activity

1. A binomial is a part of two of six pupils: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

2. The president can be chosen in six ways, then the secretary from among the other five students: $6 \times 5 = 30$ distributions.

3. Each intervention may be entrusted to one of the six students, including one already chosen student. three successive choices give $6^3 = 216$ sequences.

4. With four cards, the pairs are AB, AC, AD, BC, BD, CD : six choices. The twelve couples of roles are:

$$\begin{array}{lll} (A,B) & (A,C) & (A,D) \\ (B,A) & (B,C) & (B,D) \\ (C,A) & (C,B) & (C,D) \\ (D,A) & (D,B) & (D,C) \end{array}$$

   There are three follow-up interventions: $4^3 = 64$, for each position has four choice.

5. For a pair, the order doesn't count and you don't repeat a student. For roles, the order counts, For interventions, order counts and repetition is allowed.

E1. There are 10^4 Those with no 0 have nine choices for each of the four Positions 9^4 codes. As a complement, the requested number is $10^4 - 9^4 = 3439$.

E2. There are ten choices for gold, then nine for silver and eight for bronze. gives $10 \times 9 \times 8 = 720$ podiums. Changing order changes the podium.

E3. The person imposed is already chosen. It remains to choose a part of two people from the Six others: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

E4. A path is determined by the positions of its four steps to the right among seven positions. Three more steps are then imposed. $\binom{7}{4} = 35$ paths.

E5. You have to choose two girls out of eight, then two boys out of seven. gives only one group: $\binom{8}{2}\binom{7}{2} = 28 \times 21 = 588$.

Corrected from DM9

1. Order does not count: there is $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$ commissions.

2. Two students are selected from among seven and two teachers: $\binom{7}{2}\binom{5}{2} = 21 \times 10 = 210$ commissions.

3. The 495 commissions were withdrawn. $\binom{5}{4} = 5$ commissions exclusively teachers and $\binom{7}{4} = 35$ These categories are disjunct: 495 $- 5 - 35 = 455$.

4. Each of the 210 commissions has two choices of teacher president and two choices of student secretary: 210 $\times 2 \times 2 = 840$ organisations.

5. Classify commissions by number k teachers, with $0 \leq k \leq 4$. Category k contains $\binom{5}{k}\binom{7}{4-k}$ The five categories are disjunct and cover all commissions. addition,

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4} = 495.$$

Corrected work in the manual or an anal

One hand contains five distinct cards; the order does not intervene. $\binom{52}{5} = 2{,}598{,}960$ hands. For a hand containing the spades, this card is already chosen; the other four are chosen from among The remaining 51 cards. There are $\binom{51}{4} = 249{,}900$ hands. This result is not multiplied by 5 : the hand n no first or last position.